

COMP/ISYS355 Group 19 Pty Ltd
Feasibility Study

Breaking News Analysis

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Introduction

Lexxe are pioneers in applications on sentiment analysis; developing sindex, a sentiment based search engine and a news application (News and Moods) that is able to identify the sentiment of articles written by over 600 news outlets in near real time, analysing a total of 30 thousand documents each day.

A key feature that is offered by existing news application is notifications for breaking news. Traditionally, these breaking news notifications have been the domain of news editorial staff who would read all the news and push only those articles they deem to be breaking.

Lexxe is looking to automate this task. Currently, News and Moods relies on user customized keywords and a fixed time periods to send out notifications, but these notifications do not perform adequately.

Problem Identification

- Current methodology requires user initiative and will not be used by majority of users.
- Notifications may not be pushed out in a timely manner
- Notification may be pushed out multiple times.
- Topic detection and tracking is an open area of research
- There is no existing dataset of news articles that has had the breaking news articles labelled appropriately.
- Breaking news is a subjective topic, and there may be significant differences in breaking news annotations among the different annotators.
- The dataset may not be diverse enough to capture significant trends and as a result may not generalise to the validation dataset.

Opportunities

- Detect and track topics in articles.
- Utilize topics and sentiment scores to identify breaking news.
- Identify news article that are significantly more important than the rest.
- Increased user engagement with relevant notification for the user.
- Processes articles in near real time.

Mandates

- System must be able to scale with user base and indexed news publications.
- Articles should not be pushed multiple times.
- Articles should be pushed in a timely manner.
- Users data must be in accordance with privacy policy and government standards.
- Follow standard data science practices, and isolate development and validation data.
- Minimize the influence of chance, and ensure that results are reproducible

Success Factors

System and Process Related Success Factors

The goal of the project is to improve upon the Breaking News system for the “News and Moods” app. To do this, we must be able to analyse a news article and extract certain metadata.

Gathering the articles into a single database will need to be done early on, as having each article stored in a separate xml file is inefficient for processing and indexing the data.

Correctly identify the “*breaking news indicators*”:

An example of some potential indicators:

Frequency of words over time:

- Found using negative index of words, showing for each word the documents it occurs within. If a topic is popular, more news sources will mention it.

“Breaking news” phrase extraction:

- Breaking news often uses phrases like “breaking news”, “breaking”, “more to come”. It is a good indicator in theory, however in practice some news sources may overuse the phrases. Therefore a breaking-reliability score for each news source may be needed to test this.

The goal is for a *modular approach* to the indicators. We will be able to gather data for multiple indicators and selectively choose which ones take precedence.

The team members who are data analysts will work between both teams and identify breaking news articles manually to create test data. An interface will need to be created to display the articles to the viewer in a readable format, and allow them to return their decision on if the article contains .

This is useful for creating training and test data, which can be used to understand how successful the *breaking news indicators* are. This process should begin early on in the project.

Configurable variables that can be adjusted for different “breaking news indicators” will be useful depending on the solution used. The variables would be fine tuned based on the previously created training data, and with the use of machine learning techniques.

Project Related Success Factors

Clarity: establishing a clear definition of the projects goals. The team will communicate between ourselves, the sponsor, and the other team (team 20) to further refine the goals of the project.

Task allocation / managing team resources: Team members have different backgrounds such as coding, data analytics, project management. The effective use of their specialties depends on allocating members to the task that most suits them. In this group, each of us has some background in programming, so the tasks will be shared more evenly.

Time management: Establishing achievable deadlines for work to be done, allowing breathing room for any unforeseen roadblocks during the development cycle.

Support: There will be documentation provided for the service. The sponsor requires information on how to adapt our service into the “News and Moods” system, such as how to configure it, adjust key values, etc. The user will need an online manual about how the service

Agile Development: We will create an initial version of the system with basic functionality. More advanced capabilities will then be added iteratively over the span of the project. At each stage, the feedback from the sponsor will be vital to shaping the successive versions

Current Situation

Customer Snapshot

Lexxe provides a search engine that includes sentiment scores with the returned results from approximately 600 different sources. This is the first search engine in the world to offer this feature. A 5 star system is used where scores between 0.1 and 2.4 are rated negative, scores between 2.6 and 5.0 are rated positive and a score of 2.5 is rated neutral. The News and Moods app, designed for smartphones, extends this service by either providing push notifications for breaking news or through the use of predefined keyword searches.

Current Work Process

Web Browser

When searches are performed through a web browser, a drop down list is provided to alter the type of search required. By default, "All sentiment" is selected from the drop down list but options are provided to search for one of the 3 sentiments only or to disable the sentiment rating in the search results. An additional option is also provided to give some basic stats on the word(s) used in the search. Apart from searches on keywords, links are provided to view current news items which can be refined to display only Good News (Positive) or Bad News (Negative)

News & Moods

The News & Moods app extends the options provided through web browser searching by the means of push notifications. Push notifications are either in the form of breaking news or customised keyword searches configured through the MyNews service. For MyNews, up to 10 keyword searches can be saved. For each keyword search, a time frequency for the push notification is required with options of 1, 6, 12 or 24 hours. Searches can be further refined to only positive or negative sentiments and country (Edition) in which the results are sourced.

Benefits

Tangible Benefits

- The system will almost certainly be cheaper to run than an equivalent team of human editors, providing valuable work at lower cost
- This method will be much faster than human editors. While human editors would have to constantly watch a feed of data, then categorize it and publish it, an automated system could potentially respond in seconds.
- Due to automation, historical records are kept far better than human ones. Should the system create proper logs, it is easy to recall when and why decisions were made by the system.

Intangible Benefits

- This system reduces the potential for human error when publishing news. Errors such as typos, or accidentally publishing the wrong article are eliminated.
- This system if properly designed removes some element of human bias from news selection. Of course it's impossible to entirely eliminate, as any algorithm is tainted in some way by the bias of the programmer, there are many situations where the bias of an editor could interfere, in cases where the programmer wouldn't even imagine.
- This system removes a vector of exploitation should editors have an agenda concerning the news they intend to publish.
- Users will appreciate subscribing to a service that is reliably able to push out notifications about breaking news faster than any other service. While a news outlet may be the first to break the story, this service would be the first to post breaking news of an aggregate of all news outlets.

Alternative Solutions

Alternative 1 – Human Learning (Rule based)

Advantages

- Simple
- Does not create extra software dependencies.
- Very likely to have much better runtime performance.
- Valuable baseline for other model.
- Does not sleep.

Disadvantages

- Requires in depth domain knowledge to derive rules that are able to generalize across news sources.
- May not perform well on live data.

Alternative 2 – Machine Learning

Advantages

- Will be able to process articles significantly faster than editorial staff.
- Very likely to outperform rule based approach.
- Does not sleep.

Disadvantages

- Complex
- May require (computationally expensive) retraining periodically
- May over-fit to training data.

Alternative 3 – Extend existing code

Advantages

- The existing code can be used as a backbone for the new requirements of this project.
- Negates the need for “re-inventing the wheel”.
- Allows more time to focus on the changes required to meet the goals of the project.

Disadvantages

- The current code may not be efficient enough to support the requirements of the project.
- The methodology of the existing code may be incompatible with the proposed solution.

Alternative 4 – Editorial Team

Advantages

- Highest probability of defining news that is breaking.
- This is a proven method, there aren't any new hard problems to solve so the chance of successful deployment is much higher.
- Scales well, just hire more editors.

Disadvantages

- Costs for hiring an editorial team are high.
- The high costs of this method are ongoing, and unlikely to be reduced over time without the production of software tooling to aid human editors.
- No inherent competitive edge over competitors.

Alternative 5 – No Action

Advantages

- Lowest cost option

Disadvantages

- No working solution

Recommended Solution

Human learning or the rule based approach consists of data scientists analyzing the data to derive specific rules such as if the article is published by example.com and contains the phrases “Breaking News” and “more to come on example.com”. This approach is probably the solution which would take the least amount of time to develop a relatively simple solution that performs reasonably. However, further time spent on this approach is not likely to increase the predictive performance.

Machine learning is the most technically challenging, but potentially the highest reward solution. First we would select a set of features that we find to be good indicators of whether news is breaking or not, and automate the extraction of these features of scraped data. We would then feed these features into a machine learning classifier in an attempt to accurately predict breaking news. This solution will scale to very large amounts of input, and should grow more accurate with larger volumes of data. Machine learning also has the benefit of low ongoing costs, similar to the Human learning approach.

Hiring an editorial team would be the safest solution, however ongoing costs such as wages make this approach infeasible in practice, and as such is low on our list of priorities.

Although we can initially save time utilising the existing code, there does appear to be some compatibility issues with the approach we plan to take. The existing structure of the code can be rewritten to be more efficient and methodical in its layout. This will better facilitate the remaining code required to meet the needs of this project.

No action would certainly be the easiest solution, however we do not believe that existing code is in a finished state, and as such, no action would provide no workable solution, and as such should not be considered.

Ultimately, **we recommend using the machine learning approach**; however, there is good reason to spend a portion of the time developing a very simple version of the rule based approach to serve as an appropriate baseline which we can use to compare the performance of the machine learning based approach.
